

Practice Exercises

1 Managing Services with systemctl

- List all active services on the system.
- Check if the sshd service is running.
- Start and enable the firewalld service to run at boot.
- Restart crond and check its status.
- Stop the cups service and disable it from starting at boot.

2 Networking Basics & Configuration

- Display all network interfaces and their assigned IP addresses.
- Show the system's default gateway.
- Use ss to list all listening TCP ports.
- Retrieve the hostname and change it to student-PC.
- Verify that the hostname change was applied.

3 Network Manager & nmcli

- List all active network connections.
- Display detailed information about a specific connection.
- Add a new static network connection with IP 192.168.2.100/24.
- Modify an existing connection to use 8.8.8.8 as the primary DNS server.
- Bring the modified connection up.

Week 6 practical/lab answers

The following answers correspond directly to the practice exercises on the final slide.

Practical 1 – Managing services with **systemctl**

Question 1: List all active services on the system.

```
systemctl list-units --type=service
```

Expected result: A table showing active service units.

For all services, including inactive ones:

```
systemctl list-units --type=service --all
```

Question 2: Check whether the **sshd service is running.**

```
systemctl is-active sshd.service
```

Expected result when running:

```
active
```

For full details:

```
systemctl status sshd.service
```

Question 3: Start and enable `firewalld` to run at boot.

Start it now:

```
sudo systemctl start firewalld.service
```

Enable it at boot:

```
sudo systemctl enable firewalld.service
```

Verify:

```
systemctl is-active firewalld.service
systemctl is-enabled firewalld.service
```

Expected:

```
active
enabled
```

Combined alternative:

```
sudo systemctl enable --now firewalld.service
```

Question 4: Restart `crond` and check its status.

```
sudo systemctl restart crond.service
systemctl status crond.service
```

Quick verification:

```
systemctl is-active crond.service
```

Expected:

```
active
```

Question 5: Stop `cups` and disable it from starting at boot.

```
sudo systemctl stop cups.service
sudo systemctl disable cups.service
```

Verify:

```
systemctl is-active cups.service  
systemctl is-enabled cups.service
```

Expected:

```
inactive  
disabled
```

Combined alternative:

```
sudo systemctl disable --now cups.service
```

If the service is not installed, the system may report that `cups.service` could not be found.

Practical 2 – Networking basics and configuration

Question 1: Display all network interfaces and assigned IP addresses.

```
ip address
```

Short form:

```
ip a
```

Expected result: Interfaces such as `lo`, `enp0s3` and possibly `enp0s8`, together with their addresses.

Question 2: Show the system's default gateway.

```
ip route
```

Look for:

```
default via GATEWAY_IP dev INTERFACE
```

To show only the default route:

```
ip route | grep default
```

Question 3: Use `ss` to list all listening TCP ports.

```
ss -tln
```

Include processes:

```
sudo ss -tlnp
```

Expected result: Listening TCP sockets such as SSH on port 22.

Question 4: Retrieve the hostname and change it to `student-PC`.

Retrieve it:

```
hostname
```

Change it:

```
sudo hostnamectl set-hostname student-PC
```

Question 5: Verify that the hostname change was applied.

```
hostname
```

```
hostnamectl
```

Expected hostname:

```
student-PC
```

Practical 3 – NetworkManager and `nmcli`

Question 1: List all active network connections.

```
nmcli connection show --active
```

Expected result: Active connection names, UUIDs, types and devices.

Question 2: Display detailed information about a specific connection.

First identify its name:

```
nmcli connection show
```

Then:

```
nmcli connection show "Wired connection 1"
```

Replace **"Wired connection 1"** with the actual connection name shown on your VM.

Question 3: Add a static connection with IP **192.168.2.100/24.**

First identify the available interface:

```
nmcli device status
```

Assuming the interface is **enp0s8**:

```
sudo nmcli connection add \  
type ethernet \  
con-name lab-static \  
ifname enp0s8 \  
ipv4.method manual \  
ipv4.addresses 192.168.2.100/24 \  
ipv4.gateway 192.168.2.1
```

Verify that it was created:

```
nmcli connection show lab-static
```

Do not blindly use **enp0s8**; use the interface name displayed on your machine.

Question 4: Modify an existing connection to use **8.8.8.8 as the primary DNS server.**

```
sudo nmcli connection modify lab-static ipv4.dns 8.8.8.8
```

Verify the saved value:

```
nmcli connection show lab-static | grep ipv4.dns
```

Expected:

```
ipv4.dns: 8.8.8.8
```

Question 5: Bring the modified connection up.

```
sudo nmcli connection up lab-static
```

Verify:

```
nmcli connection show --active
```

```
ip address show enp0s8
```

```
ip route
```

Expected results:

- `lab-static` appears as active.
 - `enp0s8` has `192.168.2.100/24`.
 - A route through `192.168.2.1` may appear.
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Complete Week 6 lab workflow

This combines both official lab topics into one clean practice sequence.

Part A – Service management

```
systemctl list-units --type=service
```

```
systemctl status sshd.service
```

```
systemctl is-active sshd.service
```

```
systemctl is-enabled sshd.service
```

Start and enable the firewall:

```
sudo systemctl start firewalld.service
```

```
sudo systemctl enable firewalld.service
```

Verify:

```
systemctl is-active firewalld.service
```

```
systemctl is-enabled firewalld.service
```

Restart the scheduler:

```
sudo systemctl restart crond.service
```

```
systemctl status crond.service
```

Stop and disable printing:

```
sudo systemctl stop cups.service
```

```
sudo systemctl disable cups.service
```

Verify:

```
systemctl is-active cups.service  
systemctl is-enabled cups.service
```

Part B – Inspect the network

```
ip address  
ip route  
nmcli device status  
nmcli connection show  
nmcli connection show --active  
ss -tln  
hostnamectl
```

Record:

- Interface name
 - Current IPv4 address
 - Default gateway
 - Active connection name
 - Current hostname
 - Listening ports
-

Part C – Configure a static connection

Assuming the unused interface is `enp0s8`:

```
sudo nmcli connection add \  
type ethernet \  
con-name lab-static \  
ifname enp0s8 \  
ipv4.method manual \  
ipv4.addresses 192.168.2.100/24 \  
ipv4.gateway 192.168.2.1
```

Configure DNS and automatic startup:

```
sudo nmcli connection modify lab-static ipv4.dns 8.8.8.8  
sudo nmcli connection modify lab-static connection.autoconnect yes
```

Activate:

```
sudo nmcli connection up lab-static
```

Verify:

```
nmcli connection show --active
nmcli connection show lab-static
ip address show enp0s8
ip route
```

Test the gateway:

```
ping -c 4 192.168.2.1
```

This ping only succeeds when an actual gateway exists at that address on the connected network.

Part D – Clean up the practice connection

Deactivate:

```
sudo nmcli connection down lab-static
```

Delete:

```
sudo nmcli connection delete lab-static
```

Verify:

```
nmcli connection show
```

Common Week 6 mistakes

Confusing **start** with **enable**

```
sudo systemctl start sshd
```

starts SSH now but does not necessarily start it after reboot.

```
sudo systemctl enable sshd
```

configures startup at boot but does not necessarily start it immediately.

Use both when required.

Assuming enabled means running

Always check both:

```
systemctl is-active SERVICE  
systemctl is-enabled SERVICE
```

Using the wrong service name

Possible correct format:

```
systemctl status sshd.service
```

Not:

```
systemctl status ssh
```

The exact service name may vary, so check:

```
systemctl list-unit-files --type=service | grep ssh
```

Using the wrong network interface

Do not assume it is `eth0` or `enp0s8`.

Check:

```
nmcli device status  
ip link
```

Forgetting to activate an edited connection

After:

```
nmcli connection modify ...
```

apply it with:

```
sudo nmcli connection up CONNECTION_NAME
```

Confusing device names and connection names

Example:

Device: enp0s3

Connection: Wired connection 1

Commands such as `ip address show` normally use the device name.

Commands such as `nmcli connection modify` use the connection name.

Breaking the active connection

Running:

```
sudo nmcli connection down "Wired connection 1"
```

may disconnect the VM or terminate a remote SSH session.

Use an unused adapter for practice where possible.

Duplicate connection name

This fails when `lab-static` already exists:

```
nmcli connection add con-name lab-static ...
```

Check and remove the old profile:

```
nmcli connection show
```

```
sudo nmcli connection delete lab-static
```

Then create it again.

Week 6 essential commands

Command	Purpose
<code>systemctl list-units --type=service</code>	List active services
<code>systemctl status SERVICE</code>	View detailed status

<code>systemctl is-active SERVICE</code>	Check whether running
<code>systemctl is-enabled SERVICE</code>	Check boot configuration
<code>systemctl start SERVICE</code>	Start now
<code>systemctl stop SERVICE</code>	Stop now
<code>systemctl restart SERVICE</code>	Restart
<code>systemctl reload SERVICE</code>	Reload configuration
<code>systemctl enable SERVICE</code>	Start at boot
<code>systemctl disable SERVICE</code>	Do not start at boot
<code>ip address</code>	Display interfaces and IP addresses
<code>ip route</code>	Display routing table
<code>ss -tln</code>	Display listening TCP ports
<code>hostnamectl</code>	View or change hostname
<code>nmcli device status</code>	Display network devices
<code>nmcli connection show</code>	Display connection profiles
<code>nmcli connection add</code>	Create a connection
<code>nmcli connection modify</code>	Edit a connection
<code>nmcli connection up</code>	Activate a connection
<code>nmcli connection down</code>	Deactivate a connection
<code>nmcli connection delete</code>	Delete a connection